Experiment-7 Conservation of Linear Momentum

In this study, we will investigate the conservation of linear momentum of the system of two pucks moving on an air table. The system is isolated from its environment (that is, no external force from an object outside the system causes energy changes in the system). The air table, being horizontal and with the friction almost eliminated, will produce no net external forces on the pucks placed on it. So, we expect the total momentum of the pucks to be conserved.

Theory

The momentum (**p**) of an object is defined as the product of its mass and its velocity:

$$\vec{p} = m\vec{v} \tag{24}$$

The direction of the momentum is the direction of its velocity and the **magnitude** is:

$$p = mv \tag{25}$$

When the net external force on a system of objects is zero, the total momentum remains constant. This is the law of the **conservation of momentum** and the total momentum of an *isolated system* of objects remains constant. The law of conservation of momentum is very useful in dealing with collisions.

If we consider a particle of constant mass (m), we can write Newton's second law for this particle as:

$$\Sigma \vec{F} = m\frac{d\vec{v}}{dt} = \frac{d}{dt}(m\vec{v}) \tag{26}$$

Thus, Newton's second law says that the net force acting on a particle equals to the time rate change of the combination, $m\vec{v}$ (the product of the particle's mass and velocity). This combination is called as the momentum or **linear momentum** of the particle.

If we substitute the definition of momentum into the Equation-(26), we get **Newton's second law** in terms of momentum:

$$\Sigma\vec{F} = \frac{d\vec{p}}{dt} \tag{27}$$

According to Equation-(**27**), the net force (vector sum of all the forces) acting on a particle equals to the time rate change of the particle's momentum. Since momentum is a vector quantity with the same direction as the particle's velocity, we must express the momentum of a particle in terms of its components.

If the particle has the velocity components (v_x, v_y), then its momentum components will be:

$$p_x = mv_x \tag{28}$$

$$p_y = mv_y \tag{29}$$

If there are no external forces (the net external force on a system is zero), the total momentum of the system (the **vector sum** of the momentum of the individual particles that make up the system) is constant or conserved. Each components of the total momentum is separately conserved.

Remember that in **any collision** in which external forces can be neglected, momentum is conserved and the total momentum before equals to the total momentum after. **Only** in **elastic collisions**, the total **kinetic energy** before equals to the total the kinetic energy after. So, in an **elastic collision** between two bodies, the initial and final relative velocities have the same magnitude. The kinetic energy of a particle is a scalar quantity. It depends on only the particle's mass and speed, not its direction of motion.

Collisions in which kinetic energy is not conserved are said to be **inelastic** collisions. The **kinetic energy** that is lost is changed into other forms of energy, often **thermal energy**, so that the total energy is always conserved. Therefore, we do need to remember that even when the kinetic energy is not conserved, the total energy is always conserved.

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For a system with the two pucks, the total momentum (P_t) before the collision will be the same as the total momentum after the collision if the friction can be ignored. By denoting **initial** momentums by the subscript "***i*** " and **final** momentums by the "f ", **the vector equation** for the principle of the conservation of momentum is given by:

$$\vec{P}_{t,i} = \vec{P}_{t,f} \tag{30}$$

$$\vec{P}_{1,i} + \vec{P}_{2,i} = \vec{P}_{1,f} + \vec{P}_{2,f} \tag{31}$$

Let the velocities of the two pucks be denoted as:

- $v_{1,i}$ and $v_{2,i}$ before the collision as the **initial** values and let the velocities after the collision be:
- $v_{1,f}$ and, $v_{2,f}$ as the **final** values.

Then, **the magnitude** of the momentum of the each puck before and after the collision will be:

$$P_{1,i} = mv_{1,i} \tag{32}$$

$$P_{2,i} = mv_{2,i} \tag{33}$$

$$P_{1,f} = mv_{1,f} \tag{34}$$

$$P_{2,f} = mv_{2,f} \tag{35}$$

Now, we can analyze the initial and final velocities of a system with the two particles (puck-A and B). If we define:

$\vec{v}_{1,i}$: The momentum of puck-A, and

$\vec{v}_{2,i}$: The momentum of puck-B just before the collision,

then the total **momentum** of the two pucks before the collisions is the **vector sum**:

$$m_1\vec{v}_{1,i} + m_2\vec{v}_{2,i} \quad \text{(Vector sum)} \tag{36}$$

Immediately after the collision, the pucks have a different velocity and momentum. The total momentum after the collision is the **vector sum**:

$$m_1\vec{v}_{1,f} + m_2\vec{v}_{2,f} \tag{37}$$

The total vector **momentum** of the system of two colliding balls is **conserved** (it stays constant):

Momentum before = Momentum after

$$m_1\vec{v}_{1,i} + m_2\vec{v}_{2,i} = m_1\vec{v}_{1,f} + m_2\vec{v}_{2,f} \tag{38}$$

or,

Momentum before = Momentum after

$$m_A\vec{v}_A + m_B\vec{v}_B = m_A\vec{v}'_A + m_B\vec{v}'_B \tag{39}$$

where,

$v_{1,i}$: The velocity of puck-**A** **before** collision (or, v_A),

$v_{2,i}$: The velocity of puck-**B** **before** collision (v_B),

$v_{1,f}$: Final velocity of puck-**A** after collision (v'_A),

$v_{2,f}$: Final velocity of puck-**B** after collision (v'_B).

When the masses of the two pucks are identical ($m=m_1=m_2$), the above relation representing the conservation of the momentum gives the velocity vector relationship:

Momentum before = Momentum after

$$\vec{v}_{1,i} + \vec{v}_{2,i} = \vec{v}_{1,f} + \vec{v}_{2,f} \tag{40}$$

or,

Momentum before = Momentum after

$$\vec{v}_A + \vec{v}_B = \vec{v}'_A + \vec{v}'_B \tag{41}$$

Conservation of the momentum is always valid as long as **no** external forces act. Total momentum is conserved for the two object system in both **elastic** and completely **inelastic collisions**. So, the sum of the initial momentum of the two objects is always equal to the sum of their final momentum.

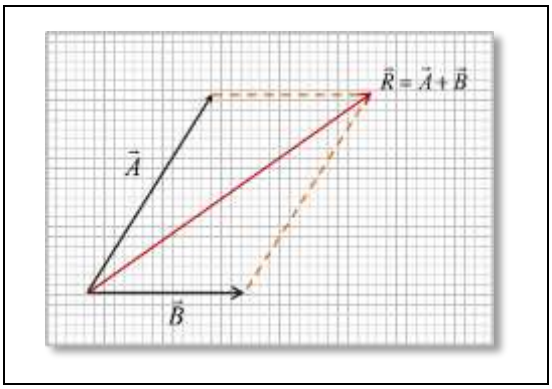


Figure-11: The way to add two vectors by a parallelogram.

When you apply the **conservation of momentum** to a system, remember that momentum is a vector quantity, so you must use **vector addition** to determine the total momentum of the system. To compute the total momentum of the system, we need to add two velocity vectors and to represent the magnitude of the velocity vector. Note that:

- The **magnitude** of a given vector in a scaled vector diagram is depicted by the length of the arrow.
- The arrow is drawn a precise length according to a chosen scale and should have has an obvious arrowhead.

For example, we have **a velocity vector** with a magnitude of 20cm/s. Then, if the scale used for constructing the diagram is chosen to be 1cm=5cm/s, the **vector arrow** of the velocity vector is drawn with a length of $4cm$, such that:

- $4cm\times[(5cm/s)/1cm]=20cm/s$

Therefore, the graphical techniques of vector addition involve drawing accurate scale diagrams to denote individual vectors and their resultants. The parallelogram method is a graphical technique of finding the resultant of two vectors. The way to add two velocity vectors:

- $\vec{A}=\vec{v}_{1,i}$ and,
- $\vec{B}=v_{2,i}$

is to construct a parallelogram as shown geometrically in the Figure-(**11**).

According to the parallelogram law, if two vector quantities are represented by two adjacent sides or a parallelogram, then the diagonal of the parallelogram will be equal to the resultant vector (vector sum), $\vec{R}$ of these two vectors. The resultant is the diagonal of the parallelogram. The magnitude can be determined from the length of its arrow using the scale.

To find the sum of the two vectors geometrically:

1. *Place vectors of:*
   - $\vec{A}$ and $\vec{B}$

   such that their initial points ***coincide****.*
2. *After that, complete a parallelogram that has* $\vec{A}$ *and* $\vec{B}$ *as two sides.*
3. ***The diagonal of the parallelogram*** *with the same initial points as* $\vec{A}$ *and* $\vec{B}$ *is the sum of* $\vec{A}+\vec{B}$.
4. *The* ***vector sum*** $\vec{A}+\vec{B}$ *is called the resultant of vectors* $\vec{A}$ *and* $\vec{B}$.

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In this laboratory, we are also going to investigate the conservation of the kinetic energy of the pucks in the course of the collision.

In an **elastic collision**, the **kinetic energy** of the each colliding object may change, but the total kinetic energy of the system does **not** change. However, in an **inelastic collision** of two objects, the kinetic energy of the two-body system is not conserved. If the objects stick together the collision is a completely inelastic collision and the objects have the same final velocity.

Let's look at an **elastic collision** between two bodies (puck-A and puck-B). To investigate the **conservation of the kinetic energy** of the pucks experimentally, remember the definition of the kinetic energy (K) of an object of mass (m) and linear velocity (v):

$$K = \frac{1}{2}mv^2 \qquad \text{(Kinetic Energy)} \qquad (42)$$

Therefore, the **total kinetic energy** (**K**) of the two-puck system before an **elastic collision** will be:

$$K = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 \qquad \text{Elastic (Before Collision)} \qquad (43)$$

and after the collision, the total kinetic energy:

$$K' = \frac{1}{2}mv_A'^2 + \frac{1}{2}mv_B'^2 \qquad \text{Elastic (After Collision)} \qquad (44)$$

The kinetic energy is conserved in an **elastic collision** such that:

$$\boxed{K = K'} \qquad \text{(Conservation of Kinetic Energy)} \qquad (45)$$

In the completely **inelastic collision**, momentum will obviously be conserved too, since the system is still an isolated one. **But**, the two pucks in the collision will stick together and form a single object with mass **2*m*** that moves with velocity $\vec{v}'$.

The total kinetic energy after the collision will be:

$$K' = \frac{1}{2}(2m)v'^2 = mv'^2 \qquad \text{Inelastic (After Collision)} \qquad (46)$$

In the **elastic collision**, the total kinetic energy should be the same after the collision as it was before the collision. However, it is not conserved in the completely **inelastic collision**. The fractional loss of the kinetic energy is defined as:

$$Fractional\ \ loss = \frac{K - K'}{K} \qquad (47)$$

We can also define the percentage fractional loss in the kinetic energy as:

$$\boxed{\%loss = \frac{K - K'}{K} \times 100} \qquad \text{(Fractional loss)} \qquad (48)$$

We can calculate the percentage fractional loss of the kinetic energy in this experiment by using Equation-(48).

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7.1. Experimental Procedures

This experiment will be carried on a horizontal leveled air table. We will investigate the conservation of momentum for two pucks moving on the frictionless horizontal air table and we will assume there is no net external force on the system. The two pucks will be allowed to collide and then their total momentum before and after the collision will be measured. The details of the experimental procedures are described below.

1. Adjust the legs of the air table for a horizontal leveled air table.

2. Activate only the pump (compressor) switch and push the two pucks diagonally towards each other across the air table.

   2.1. Repeat this procedure several times until you get a satisfactory collision.
   2.2. Don't push the two pucks **too** slowly or too fast, but push them such that they move with a moderate speed.
   2.3. You will find the best speed after several experimental test observations.

3. Now, turn on the spark timer and choose the spark timer frequency as:
   - $f = 20Hz$.

4. Activate the compressor and spark timer switches in the same time, and then push the **two pucks** diagonally to get a collision nearly at the center of the table.

5. Release the two switches when they complete their motion after the collision.

6. **Turn off the spark-timer and then remove the pucks from the air table for safety.**

7. Now, remove the data sheet on the air table to examine the dots produced.

   7.1. The dots of the two pucks should look like those in the Figure-(12a).

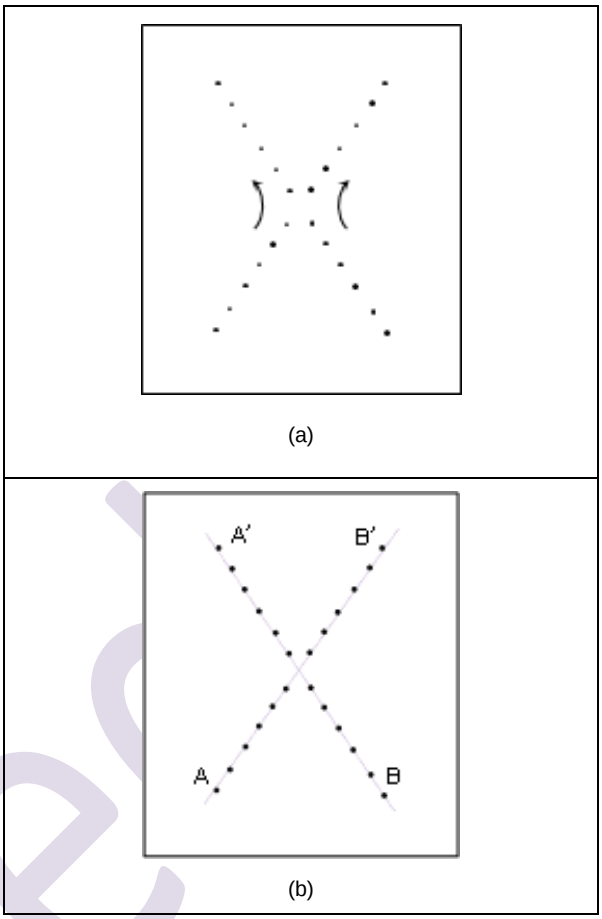


Figure-12: The data points in the elastic collision of the two pucks on a leveled air table (a).The two trajectories of the pucks by **A** and **B** before collision and **A′** and **B′** after the collision (b).

   7.2. Number the dots produced by each puck on the data sheet as **0, 1, 2, 3, 4, 5…10**. You do **not** need to start from the first dot.
   7.3. Label the two trajectories produced by the pucks as **A** and **B** before the collision and then **A′** and **B′** after the collision (Figure-12b).

8. Find the velocity (**v**) of each puck before and after the collision by measuring the length of two or three **intervals** of each trajectory with a ruler and dividing by time.

Since the spark timer frequency is set to f=20Hz, each puck on the table marks 20 dots on the experiment paper in one second, so the time **interval** Δt between two successive dots is given by T=1/20=0.05 second.

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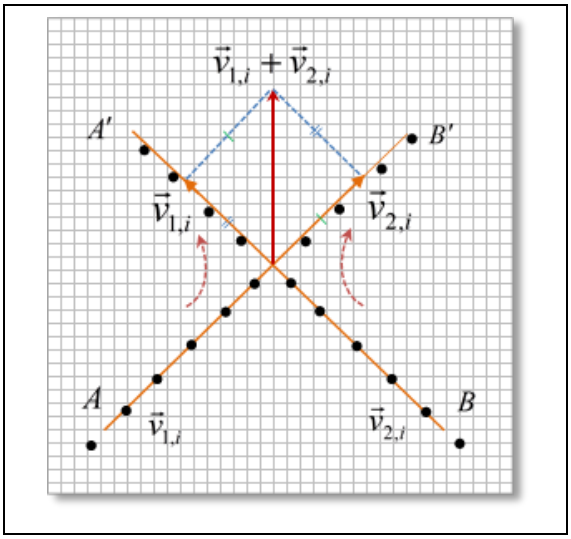


Figure-13: Vector sum given geometrically by constructing a parallelogram.

9. Find the **vector sum** $\vec{v}_{1,i}+\vec{v}_{2,i}$.

9.1. To find vector sum $\vec{v}_{1,i}+\vec{v}_{2,i}$, extend the trajectories **A** and **B** until they intersect on the data sheet as seen in the Figure-(**13**).

9.2. Then, starting from the intersection point, **draw** vectors along the directions of the velocity **vectors** $\vec{v}_{1,i}$ and $\vec{v}_{2,i}$ and with lengths proportional to the **magnitudes** of these velocities $\left|\vec{v}_{1,i}\right|$ and $\left|\vec{v}_{2,i}\right|$.

9.3. You can for instance draw a **1cm** vector to represent a **velocity** of **10cm/s** on your data sheet or on a graph paper having a **scale with 1cm=10cm/s**.

9.4. The **length** of the each vector represents the magnitude and it points in the direction of the velocity.

9.5. As shown in the Figure-(**13**), **add** the two velocity vectors by using parallelogram method **to find** the vector sum (**resultant vector**) of these velocities as:

- $\vec{v}_{1,i}+\vec{v}_{2,i}$ and its magnitude as:
- $\left|\vec{v}_{1,i}+\vec{v}_{2,i}\right|$.

9.6. Record the results in the Table-(7).

10. Repeat the same procedure to find the vector sum of $\vec{v}_{1,f}+\vec{v}_{2,f}$ and the corresponding magnitude.

11. Find the total kinetic energy of the two pucks before and after the collision. Is the kinetic energy conserved?.

7.2. LABORATORY REPORT

Conservation of Linear Momentum

Table-7: Conservation of the linear momentum.

$v_{1,i}\,(m/s)=$	. . . . .
$v_{2,i}\,(m/s)=$	. . . . .
$v_{1,f}\,(m/s)=$	. . . . .
$v_{2,f}\,(m/s)=$	. . . . .
$\left\|\vec{v}_{1,i}+\vec{v}_{2,i}\right\|=$	. . . . .
$\left\|\vec{v}_{1,f}+\vec{v}_{2,f}\right\|=$	. . . . .
$K=$	. . . . .
$K'=$	. . . . .

Conclusions and Questions

- How is the **total momentum** of the system after the collision compared with that before the collision?. Discuss the conservation of the linear momentum by using the data values.
- Write the expression for the total **kinetic energy** of the system both before and after the collision.
- Is the **kinetic energy** of the system conserved in the **elastic collision** of two objects?.
- Is the **kinetic energy** conserved in **completely inelastic collisions**?.
- In the **elastic** or completely **inelastic collisions**, is the **total momentum** of the system always conserved before and after the collisions?.

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